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Environment:

- Application: Todo Manager REST API
- JAR release: runTodoManagerRestAPI-1.5.5.jar
- API model version: v3
- OS: Windows 11
- Host: localhost:4567
- Testing tool: Windows PowerShell
- Test scope: Categories API
- Duration: 45 mins

Session notes:

1. Retrieving categories and creating categories:
    - Comment: they work as intended.
2. Retrieving categories' headers:
    - Comment: the documentation suggests that `HEAD /categories` should return a header for each category instance. However, only one is returned when there are many categories, so the documentation is inaccurate.
    - Screenshot: verifying_category_collection_headers.png
3. Retrieving information for GET, HEAD, and DELETE by ID:
    - Comment: they work as intended.
4. Updating the categories with POST and PUT:
    - Comment: both POST and PUT update the specified category fields. However, PUT clears fields omitted from the request body, whereas POST preserves them. This behavior is not explained in the API documentation.
    - Screenshot: POST_vs_PUT_category_update.png
5. Retrieving, creating and getting headers for category-project relationship:
    - Comment: the tested endpoints behaved as expected. `POST /categories/:id/projects` creates a relationship between an existing category and project. Repeating the request with the same project ID did not produce a duplicate in the relationship list. The POST request also did not modify the existing project's attributes.
6. Deleting category-project relationship:
    - Comment: this works as intended.
7. Retrieving, creating, getting headers, and deleting for category-todos relationship:
    - Comment: everything works as intended. Repeating POST did not create a duplicate category-todo relationship.

Summary of findings:

- Category creation, retrieval, updating, and deletion generally work as expected.
- Category-project and category-todo relationships can be created, retrieved, and deleted successfully.
- Repeated relationship creation does not produce duplicate relationships.

- Creating relationships does not unexpectedly modify the associated resources' attributes.

Concerns:

- `HEAD /categories` appears inconsistent with its documentation regarding returned headers.
- `POST` and `PUT` update categories differently when fields are omitted and this was never documented. This may cause accidental data loss.

New Testing Ideas:

- Test `POST` and `PUT` with empty bodies and partially omitted fields.
- Test category creation with missing, invalid, or duplicate field values.
- Test requests with an empty ID field.
- Test IDs containing non-numeric or special characters, such as `abc`, `!@#`, or spaces
- Test deleting the same category twice.
- Test malformed JSON and XML request bodies.